

Energy–Latency Aware Digital Twin-Slice Co-Optimization

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1 Network Entities

- A 6G gNB serves users through network slices ($\mathcal{S} = \{1, \dots, S\}$) (e.g., eMBB/URLLC/mMTC).
- Digital-twin services run on edge DT servers ($\mathcal{I} = \{1, \dots, I\}$) (each hosts one or more DT instances).
- Users ($\mathcal{U} = \{1, \dots, U\}$) request DT-assisted services (robot control, XR telepresence, vehicular DT sync, smart factory monitoring).

2 Practical User Requirements (per user u)

- Data demand (payload) d_u [bits] that must be sent to the edge for DT inference/synchronization.
- Latency deadline τ_u [s].
- Reliability target ϵ_u (so required success prob. $\geq 1 - \epsilon_u$).
- Compute intensity α_u [cycles/bit] (or per-DT compute coefficient; I'll keep it user-specific for generality).

3 Slice and DT Capacities

- Slice bandwidth budget B_s [Hz] and usable fraction $\beta_s \in (0, 1)$.
- Slice backhaul/fronthaul budget H_s [bits/s] (or “traffic budget”).
- DT compute capacity C_i [cycles/s].

4 Radio + Propagation Parameters

- Spectral efficiency $\eta_{u,s}$ [bits/s/Hz] if user u is served in slice s .
- Reliability success probability $p_{u,s}$ (slice-dependent PHY reliability proxy).
- Propagation + queueing/transport latency between slice s and DT server i : $h_{i,s}$ [s].

5 Decision Variables

5.1 Admission and Association (binary)

- $y_u \in \{0, 1\}$: 1 if user u is admitted (served), else 0.
- $x_{u,i,s} \in \{0, 1\}$: 1 if admitted user u is assigned to DT i and slice s .

5.2 Resource Allocation (continuous)

- $b_{u,s} \geq 0$: bandwidth allocated to user u in slice s [Hz].
- $f_{u,i} \geq 0$: compute cycles/s allocated to user u at DT server i [cycles/s].

6 Linking Constraints (Consistency)

$$\sum_{i \in \mathcal{I}} \sum_{s \in \mathcal{S}} x_{u,i,s} = y_u, \quad \forall u \quad (\text{C1})$$

Meaning: each admitted user chooses exactly one (DT, slice) pair.

$$0 \leq b_{u,s} \leq \left(\sum_i x_{u,i,s} \right) B_s, \quad \forall u, s \quad (\text{C2})$$

Meaning: bandwidth only if user is assigned to that slice.

$$0 \leq f_{u,i} \leq \left(\sum_s x_{u,i,s} \right) C_i, \quad \forall u, i \quad (\text{C3})$$

Meaning: compute only if user is assigned to that DT.

7 Latency and Reliability Models (Common)

End-to-end latency for an admitted user.

Transmission latency (uplink payload to edge):

$$T_u^{\text{tx}} = \sum_{s \in \mathcal{S}} \frac{d_u}{\eta_{u,s} b_{u,s} + \delta}, \quad (\text{L1})$$

where $\delta > 0$ avoids division by zero in numerical simulation.

Compute latency at DT:

$$T_u^{\text{cp}} = \sum_{i \in \mathcal{I}} \frac{\alpha_u d_u}{f_{u,i} + \delta}. \quad (\text{L2})$$

Transport latency:

$$T_u^{\text{tr}} = \sum_i \sum_s x_{u,i,s} h_{i,s}. \quad (\text{L3})$$

Total:

$$T_u = T_u^{\text{tx}} + T_u^{\text{cp}} + T_u^{\text{tr}}. \quad (\text{L4})$$

Latency QoS:

$$T_u \leq \tau_u + M(1 - y_u), \quad \forall u \quad (\text{C4})$$

Meaning: if admitted ($y_u = 1$), enforce deadline; else relaxed by big- M .

Reliability constraint

$$\sum_i \sum_s x_{u,i,s} p_{u,s} \geq (1 - \epsilon_u) y_u, \quad \forall u \quad (\text{C5})$$

Meaning: admitted users must choose a slice that meets the reliability target.

8 Slice/DT Capacity Constraints (Common)

Bandwidth budget per slice:

$$\sum_{u \in \mathcal{U}} b_{u,s} \leq \beta_s B_s, \quad \forall s \quad (\text{C6})$$

Backhaul/traffic budget per slice (one practical model):

$$\sum_{u \in \mathcal{U}} y_u d_u \leq H_s T_{\text{win}}, \quad \forall s \quad (\text{C7})$$

where T_{win} is the scheduling window duration (e.g., 1 time slot).

Meaning: total admitted traffic mapped to slice s must fit its transport budget.

(Alternative: use rate form $\sum_u y_u r_u \leq H_s$.)

Compute budget per DT server:

$$\sum_{u \in \mathcal{U}} f_{u,i} \leq C_i, \quad \forall i \quad (\text{C8})$$

Binary integrality:

$$x_{u,i,s} \in \{0, 1\}, \quad y_u \in \{0, 1\}. \quad (\text{C9})$$

9 Research Problem 1

Energy–Latency Aware Digital Twin–Slice Co-Optimization

9.1 Scenario (Detailed)

A 6G private/public network runs multiple slices. Users request DT synchronization or DT-assisted inference (e.g., XR/robotic control). When admitting many users, the edge DT servers consume significant energy (DVFS, GPU/CPU scaling) and radios consume power. The operator wants timely DT service while keeping energy cost low, under slice bandwidth/backhaul and DT compute limits, with strict URLLC-like deadlines for some users.

9.2 Energy Model (Practical)

Communication energy for user u (uplink):

$$E_u^{\text{tx}} = \sum_s P_{u,s}^{\text{tx}} T_{u,s}^{\text{tx}} = \sum_s P_{u,s}^{\text{tx}} \frac{d_u}{\eta_{u,s} b_{u,s} + \delta} \quad (\text{E1})$$

Meaning: energy = power $\times$ transmit duration (slice-dependent power can reflect MCS/reliability).

Computation energy (DVFS-style) at DT server i :

$$E_u^{\text{cp}} = \sum_i \kappa_i (f_{u,i})^2 T_{u,i}^{\text{cp}} = \sum_i \kappa_i (f_{u,i})^2 \frac{\alpha_u d_u}{f_{u,i} + \delta} \quad (\text{E2})$$

Meaning: common edge model where dynamic power $\propto f^2$ and energy scales with runtime.

Total energy:

$$E_u = E_u^{\text{tx}} + E_u^{\text{cp}}. \quad (\text{E3})$$

9.3 Objective (Energy–Latency Co-Optimization with Admission)

A clean IEEE-friendly objective is maximize admitted users while minimizing latency+energy:

$$\max_{\{x,y,b,f\}} \sum_u y_u - \lambda_T \sum_u y_u T_u - \lambda_E \sum_u y_u E_u \quad (\text{P1-Obj})$$

Meaning: primary goal is admission; secondary penalizes latency and energy among admitted users ($\lambda_T, \lambda_E > 0$ tune trade-off).

9.4 Full Problem (Problem 1)

$$(\mathbf{P1}) \quad \max (??) \quad \text{s.t.} \quad (??) - (??), (??) - (??), (??) - (??).$$

Nature: mixed-integer, non-linear (fractional terms in latency/energy) $\Rightarrow$ NP-hard (binary assignment + knapsack-like capacities).